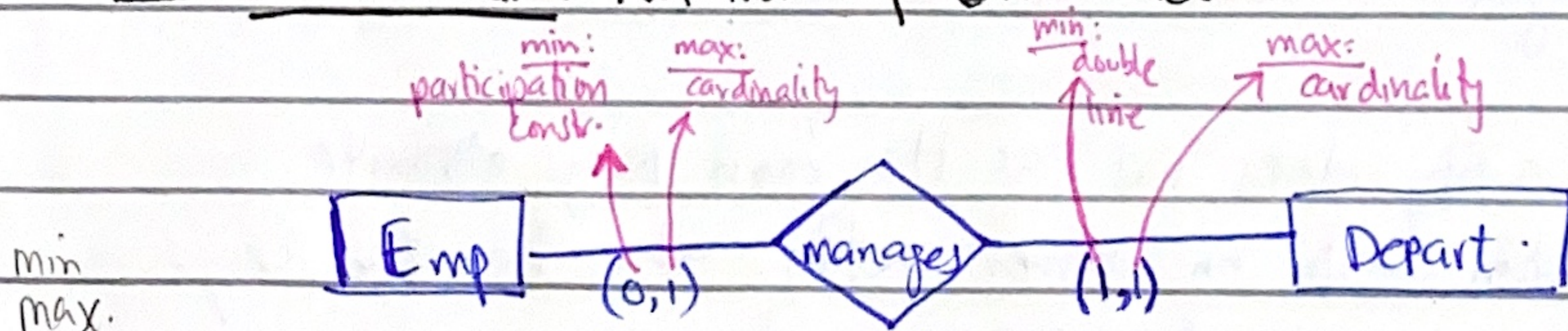


LECTURE - 20

Wednesday
15/4/26.

Min . Max : Notation of ER model.



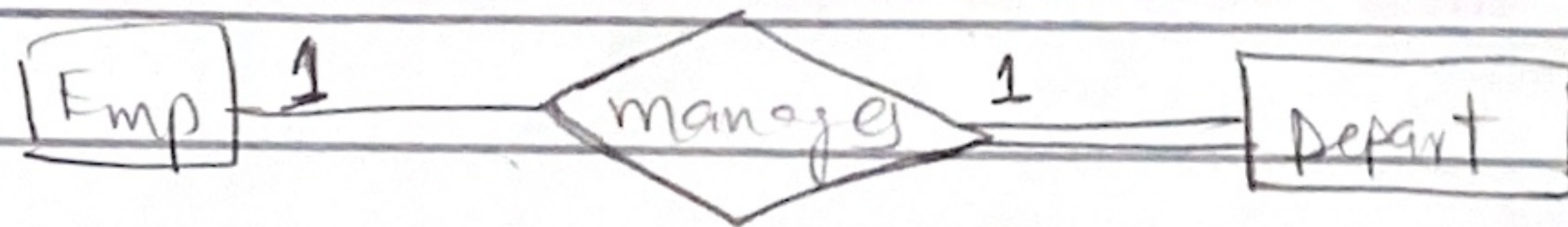
Max column shows
1 - 1 relation

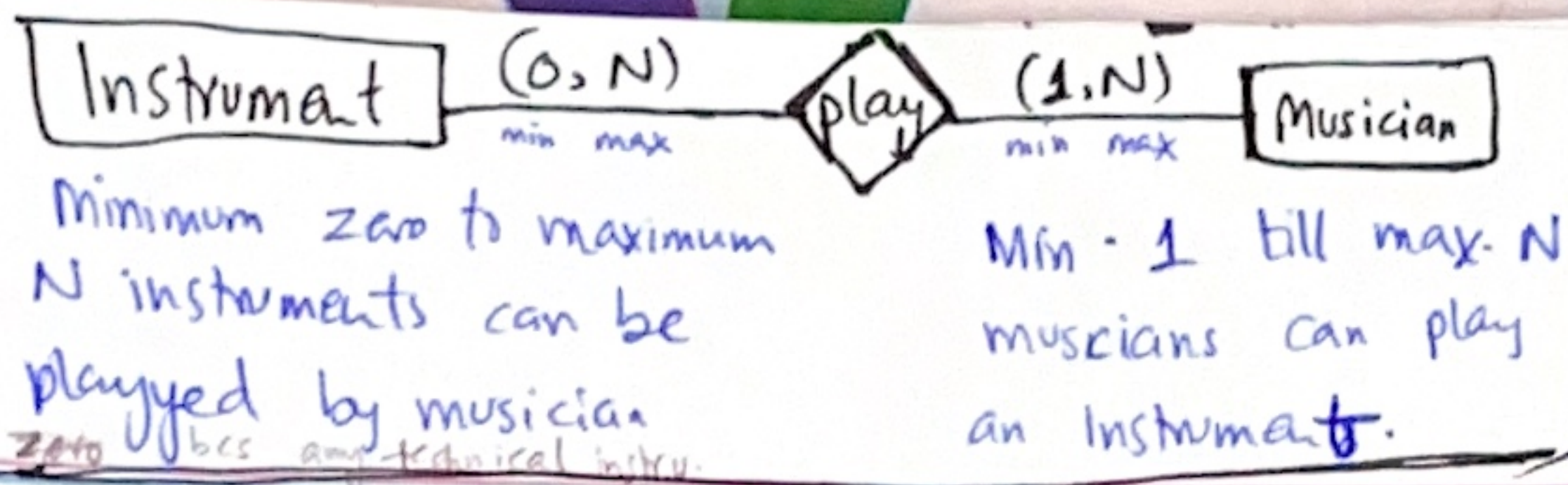
Min column shows:

Employee w single line

Depart w double line

Chen:





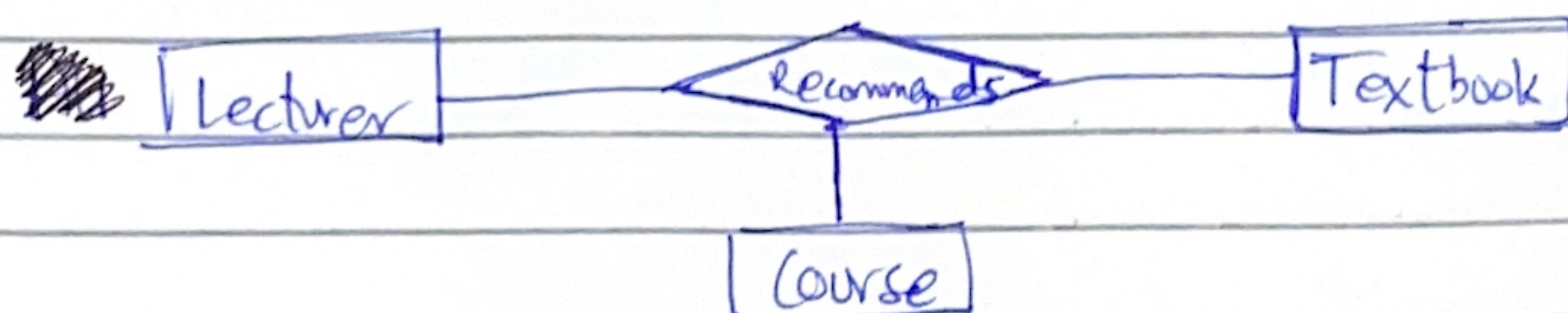
How to read Brackets in min max.

$(1, 1)$

NOTE: In min max we can do $\uparrow (3, N)$ which is a plus point over $\text{emp} \text{---} \text{Projects}$ as 'min' catches double or single line which gives extra info i.e. '3'.

This means minimum 3 projects must be with an employee whereas; Chen format only told ^{at least one} ~~projects~~ ^{must} be under employee. (NOT min. number).

Termary Relationships:



Means: Lecturer recommends Textbook for Course

e.g: Ali recommends Fundamentals of DB for Database Systems.

→ "NOT equivalent Binary Relationship" to above ternary in Slides: 18/23

⇒ Cardinality in Ternary: Slides # 21/23

* Rule: make pair of two and check it against the third like here in slides we paired project and part & checked against suppliers.

The cardinality will be towards supplier entity from diamond. (i.e. 1)

⇒ Convert Ternary to Binary: Slides # 22/23

* Here we'll introduce a weak entity i.e. SUPPLY

This is converted as the key of SUPPLY is composite key consisting of 'Sname', 'PartNo', 'Proj name' (all three) which are primary keys of its three owners & ~~they~~ connect all 3 entities which ternary was trying to display.

→ Example & Redundancy in Slide 22/23: Degree of relationship means binary, ternary etc.